

SEMICONDUCTORS

1. Semiconductors

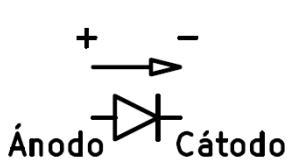
Insulating materials such as plastic or wood do not allow electric current to flow through them. Conductive materials such as copper or aluminium allow electric current to flow very easily. On the other hand, **semiconductor** materials such as **silicon** or **germanium** can behave as insulators or conductors depending on the voltage applied to them.

This behaviour can be used to manufacture circuits with very fast, electronically controlled semiconductor switches.

For semiconductors to conduct current, they must be doped with trace amounts of elements that provide positive charge carriers (boron, indium) or negative charge carriers (phosphorus, arsenic). A semiconductor doped in this way is called **P-type** (positive) or **N-type** (negative).

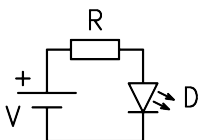
2. The diode

A diode is the simplest electronic component that can be made from semiconductor materials, and it has two terminals. Internally, it consists of a junction between a block of **P-type** silicon and a block of **N-type** silicon. This junction allows current to flow in one direction but prevents it from flowing in the opposite direction.

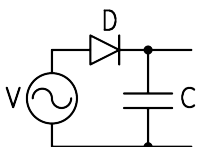


The image above shows the diode symbol, with an arrow indicating the direction in which current can flow, and the names of its two terminals. A diode conducts only when the anode has a positive voltage and the cathode has a negative voltage. On the right is a photograph of several types of diodes.

Diodes have many applications. For example, they can rectify alternating current, regulate voltages, or emit light (LEDs).



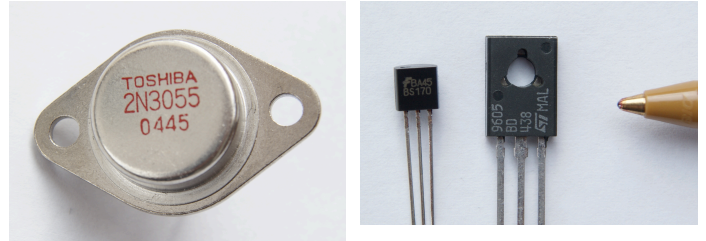
Circuit diagram of an LED connected in series with a resistor that limits the current to prevent the LED from burning out.



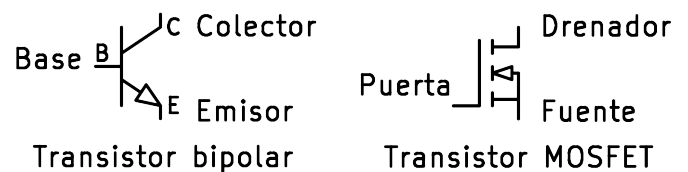
Circuit diagram of a rectifier diode that converts the alternating voltage from the mains supply into direct voltage.

3. The transistor

A transistor is a three-terminal electronic component that allows electric current to flow between two terminals according to the voltage applied to the third terminal. It acts like a voltage-controlled switch. The first silicon transistor went on sale in 1954.



Photographs of a power transistor and two small signal transistors.



Symbols for bipolar junction transistors and MOSFETs.

3.1. Transistor operating states

Depending on the control voltage applied to the transistor's base or gate, it can operate in three different states.

Cut-off: the transistor does not conduct current and behaves like an open switch.

Saturation: the transistor conducts as much current as possible and behaves like a closed switch.

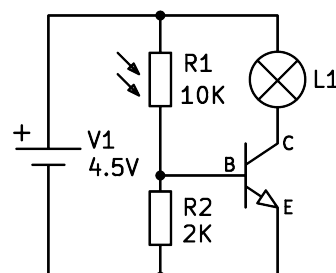
These two states are used in digital circuits such as computers, TVs, smartphones, and other electronic devices.

Linear region: the transistor conducts only part of the possible current and behaves like a resistor.

This behaviour is used in analogue circuits such as audio amplifiers.

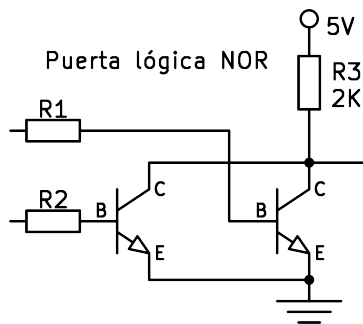
3.2. Typical circuits

Transistor amplifier. This circuit works as a light amplifier. When light shines on the LDR resistor, the current flowing through it increases.



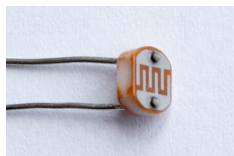
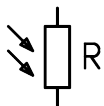
This current reaches the transistor's base, and the transistor amplifies it through the collector, switching on the connected lamp. This is an analogue circuit because the transistor operates in the linear region, behaving like a resistor controlled by the base current.

Digital transistor circuit. This circuit is a NOR logic gate built from transistors. Because the two collectors are connected in parallel, the output is high only when both inputs are low. These logic gates form the basis of digital circuits and computers.



4. LDRs

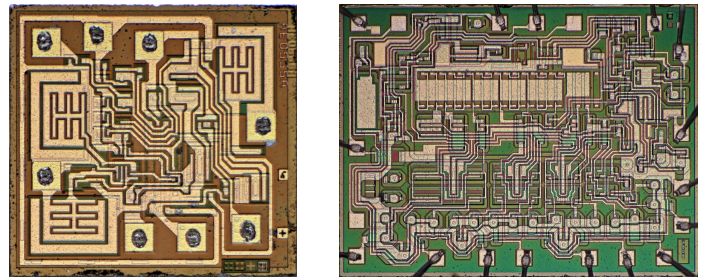
LDRs (Light-Dependent Resistors) are, as their name suggests, sensors that detect light. Their resistance decreases as the light intensity increases, allowing more current to flow when they receive more light.



Symbol and photograph of a light-dependent resistor (LDR).

5. Integrated circuits

An integrated circuit is a small piece of silicon, also called a chip, that contains a large number of electronic components inside it.



LM555 and DAC08 integrated circuits.

As technology advances, components become smaller every year, allowing more and more transistors to be integrated into a single chip. In the early 1960s, the aerospace industry began purchasing integrated circuits containing up to 100 transistors on a single chip. This reduced manufacturing costs and encouraged technological development.

By the early 1980s, chips containing 100,000 transistors were already available. In 2000, chips contained 100 million transistors, and by 2020, a single chip could contain 100 billion transistors.

This exponential growth in the number of transistors integrated into a chip, which doubles approximately every year and a half, is known as **Moore's law**. It has enabled the development of the digital society we know today, with countless smart devices, memory systems, cameras, drones, and other technologies based on these powerful integrated circuits.

EXERCISES

1. What types of materials are there, depending on how they conduct electricity? Give two examples of each type.
2. Why are semiconductors so useful?
3. What must be done for a semiconductor to conduct electric current?
4. How is a semiconductor diode constructed?
5. Draw the symbol of a semiconductor diode and label its terminals.
6. When does a diode conduct current?
7. Draw two circuit diagrams using diodes.
8. What are diodes used for?
9. What is a transistor? How many terminals does it have?
10. What operating states can a transistor have?
11. Which transistor operating states are used in analogue circuits? And in digital circuits?
12. Draw the symbols of a bipolar junction transistor and a MOSFET, labelling their terminals.
13. Draw a circuit with a transistor operating as a light amplifier.
14. Draw a NOR logic gate using transistors.
15. What is an LDR, and what do these initials stand for?
16. What is an integrated circuit, or chip?
17. When did integrated circuits begin to be manufactured, and how many transistors did they contain?
18. Draw a graph showing the number of transistors in a chip. Put the years on the X-axis and the number of transistors on the Y-axis, using an exponential scale (10, 100, 1,000, 10,000, etc.).
19. What is Moore's law?